

Socket Programming Practice Assignments

PCCSL507 – Computer Networks Laboratory

Department of Computer Science & Engineering

Objective

These practice assignments help students understand the fundamentals of TCP Socket Programming before performing advanced laboratory experiments.

Question 1 – Hello Client-Server (Beginner)

Problem Statement

Develop a TCP client-server application where:

- The client sends the message "**Hello Server**".
- The server receives the message and displays it.
- The server replies with "**Hello Client**".
- The client displays the received response.

Skills Practiced

- `socket()`
- `bind()`
- `listen()`
- `accept()`
- `connect()`
- `send()`
- `recv()`

Question 2 – Arithmetic Server

Problem Statement

Develop a TCP client-server application where:

- The client inputs two integers.
- The client sends both numbers to the server.
- The server calculates:

- Addition
 - Subtraction
 - Multiplication
 - Division
- The server sends all results back.
- The client displays the received results.

Skills Practiced

- Integer Communication
- Arithmetic Operations
- TCP Socket Programming

Question 3 – String Processing Server

Problem Statement

Develop a TCP client-server application where:

- The client inputs a string.
- The server performs the following operations:
 - Find the string length
 - Convert the string to uppercase
 - Reverse the string
- The server sends the processed results.
- The client displays all outputs.

Skills Practiced

- Character Arrays
- String Manipulation
- TCP Communication

Question 4 – Array Processing Server

Problem Statement

Develop a TCP client-server application where:

- The client inputs N integers.
- The array is sent to the server.
- The server computes:
 - Sum
 - Average
 - Maximum Element
 - Minimum Element
- The server sends the results.
- The client displays them.

Skills Practiced

- Arrays
- Loops
- TCP Socket Programming

Question 5 – Matrix Analyzer

Problem Statement

Develop a TCP client-server application where:

- The client inputs a square matrix of order N.
- The matrix is sent to the server.
- The server determines:
 - Matrix Type
 - Sum of Matrix Elements
 - Trace of Matrix
- The server sends the results.
- The client displays the received information.

Matrix Types

- Diagonal Matrix

- Upper Triangular Matrix
- Lower Triangular Matrix
- General Matrix

Skills Practiced

- Two-Dimensional Arrays
- Matrix Operations
- TCP Socket Programming

Difficulty Level

Question	Topic	Difficulty
Q1	Hello Client-Server	Beginner
Q2	Arithmetic Server	Easy
Q3	String Processing	Intermediate
Q4	Array Processing	Intermediate
Q5	Matrix Analyzer	Advanced

Submission Instructions

- Write separate Client and Server programs.
- Compile both programs successfully.
- Execute the programs using two terminals.
- Verify the output.
- Screenshot it and paste in word file and submit through github.

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